

2026 Secondary 2 Notes

Unit 06: Direct and Inverse Proportion

Name: _____ () Class: _____

Unit EUs

Students will be able to understand that:

- 1) Many variables in the real world have a **directly proportional** or **inversely proportional relationships**. Proportional relationships can be represented using tables, graphs or algebraic functions.
- 2) When modelled as algebraic functions, many problems encountered in everyday life can be solved mathematically.
- 3) The function **$y = kx$ describes a directly proportional relationship** in which y varies **directly to x** .
- 4) The function **$y = \frac{k}{x}$ describes an inversely proportional relationship** in which y varies **inversely to x** .

Unit EQs

- 1) How can proportional relationships be represented using tables, graphs or algebraic functions?
- 2) How can algebraic functions be used to model real-world situations and be used to solve everyday problems?
- 3) How can we apply $y = kx$ to represent directly proportional relationships in various contexts?
- 4) How can we apply $y = \frac{k}{x}$ to represent inversely proportional relationships in various contexts?

Unit Objectives

- Identify the type of relationship between two given variables.
- Create a suitable algebraic equation to relate the two variables.
- Apply the equation to solve future problems.
- Understand the graphical representation of the given relationship.

References

Think! Mathematics New Syllabus Mathematics 2a, 8th Edition Chapter 7

Teaching To The Big Idea ...

Lesson sequence in the unit

Student Learning Outcomes	Dimensions (Please tick the appropriate boxes)							
	FUNCTIONS	INVARIANCE	NOTATIONS	DIAGRAMS	MEASURES	EQUIVALENCE	PROPORTIONALITY	MODELS
	F	I	N	D	M	E	P	M
Direction Proportion		✓					✓	✓
Algebraic and Graphical Representation of Direct Proportion		✓		✓			✓	✓
Other Forms of Direct Proportion		✓					✓	✓
Inverse Proportion		✓					✓	✓
Algebraic and Graphical Representation of Inverse Proportion		✓		✓			✓	✓

Checklist for Self-Assessment (Please tick ✓ at the appropriate boxes)

Levels	Characteristics	Knowledge and Skills	I need help. I need to see an example.	I can do this on my own.	I can do this on my own as well as explain my solution to my classmates or teacher.
Level 0	Reproduction of facts, rules, and formulae; No explanations required	Identify graphs involving direct proportion			
Level 0	Reproduction of facts, rules, and formulae; No explanations required	Identify graphs involving inverse proportion			
Level 1	Procedural tasks without connections, Algorithmic in nature; No explanations required	Solve equations involving direct proportion.			
Level 1	Procedural tasks without connections, Algorithmic in nature; No explanations required	Solve equations involving inverse proportion.			
Level 2	Procedural tasks with connections, Algorithmic in nature; Has a "real- world" context; Explanations required	Explain the concept of inverse proportion using tables, equations and graphs			

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6.1 DIRECT PROPORTION



: Make a copy of the spreadsheet [\[Link HERE\]](#) to observe how Direct Proportion can be used in the following situation. Observe the trend as you key in the data values.

If it costs \$2000 to build a treehouse, how much does it cost to build 5 treehouses?

Complete the following table.

No. of treehouse (x)	1	2	3	4	5
Cost (\$ y)	\$2000	\$4000	\$6000	\$8000	\$10000
Rate, $\frac{y}{x}$	$\frac{2000}{1} = 2000$	2000	2000	2000	2000

- a. From the above table, what do you notice about y when x increases?

when x increases, y increases

- b. What do you notice about the rate, $\frac{y}{x}$?

The rate is a constant.

constant.

- c. Formulate an equation connecting x and y .

$$y = 2000x$$

When one quantity increases or decreases at the same *rate* as another, the two are said to be in direct proportion.

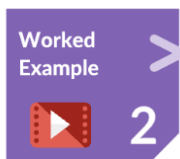
Hence, we have:

memorise!

If y is directly proportional to x , then $\frac{y}{x} = k$ or $y = kx$, where k is a constant and $k \neq 0$.



Worked Example from Textbook



Forming equation of direct proportion

If y is directly proportional to x and $y = 12$ when $x = 4$, find

- an equation connecting x and y ,
- the value of y when $x = 8$,
- the value of x when $y = 21$.

Solution

- Since y is directly proportional to x , then $y = kx$, where k is a constant.

When $x = 4$, $y = 12$,

$$12 = k \times 4$$

$$\therefore k = 3$$

$$\therefore y = 3x$$

- Method 1:**

Substitute $x = 8$ into $y = 3x$:

$$y = 3 \times 8$$

$$= 24$$

Method 2:

When $x = 8$,

$$y = 2 \times 12$$

$$= 24$$

x is doubled from $x = 4$
 y is doubled from $y = 12$

We can also use $\frac{y_2}{y_1} = \frac{x_2}{x_1}$,

$$\text{i.e. } \frac{y}{12} = \frac{8}{4}$$

$$y = 2 \times 12$$

$$= 24$$

- Substitute $y = 21$ into $y = 3x$:

$$21 = 3x$$

$$\therefore x = \frac{21}{3}$$

$$= 7$$

Attention

Since y is directly proportional to x , then $\frac{y}{x} = k$ or $y = kx$, where k is a constant and $k \neq 0$.

Reflection

In part (iii), can we use $\frac{21}{x} = \frac{12}{4}$?
 Which method do you prefer?
 How are these methods related?

Example 1:

If y is directly proportional to x and if $y = 6$ when $x = 2$, find

- an equation connecting x and y ,
- the value of y when $x = 10$,
- the value of x when $y = 9$.

(a) $\frac{y}{x} = k$, k is a constant.
 $\frac{6}{2} = k$
 $k = 3$
 $\therefore y = 3x$.

(b) when $x = 10$,
 $y = 3(10)$
 $= 30$.

(c) when $y = 9$,
 $9 = 3x$
 $x = 3$.

Example 2:

The cost of hiring a bike for a certain number of days is shown in the table.

No. of days, x	1	2	3	4
Cost, y (\$)	5	10	15	f

- (a) Find the value of f .
 (b) Find the equation connecting x and y .
 (c) Hence, find the number of days of rental if the cost is \$125.

(a) $f = 4 \times 5 = 20.$

(b) Recognise (1) direct proportion
 (2) constant.

$$y = 5x$$

$$k = \frac{y}{x}, \text{ k is a constant}$$

$$= \frac{5}{1} = 5$$

$$y = 5x$$

(c) $125 = 5x$
 $x = 25.$

Classwork (Tier A)

1. Given that y is ^{directly} proportional to x and that $y = 3$ when $x = 2$, find
 (a) y when $x = 5$, (b) x when $y = 11$

1a $k = \frac{y}{x}, \text{ k is a constant}$
 $= \frac{3}{2}.$

$$\therefore y = \frac{3}{2}x$$

when $x = 5$,

$$y = \frac{3}{2}(5)$$

$$= \frac{15}{2}$$

$$= 7.5$$

(b) when $y = 11$,

$$11 = \frac{3}{2}x$$

$$x = \frac{22}{3}$$

$$= 7\frac{1}{3}.$$

OR

Since y is directly proportional to x ,

$$k = \frac{y}{x}, \text{ k is a constant.}$$

(a) $\frac{3}{2} = \frac{y}{5}$
 $y = \frac{15}{2}$
 $= 7\frac{1}{2}$

(b) $\frac{3}{2} = \frac{11}{x}$
 $x = \frac{11(2)}{3}$
 $= 7\frac{1}{3}$

2. The extension of a spring is directly proportional to the applied force.
If a force of 20 kg stretches a particular spring to 15 cm, how far will the spring stretch when a force of 25 kg is applied?

Let the force be F

Let the extension of spring be x .

$$\begin{array}{l|l} \therefore k = \frac{F}{x}, k \text{ is a constant.} & \text{when } F = 25, \\ \frac{20}{15} & 25 = \frac{4}{3}x \\ \frac{4}{3} & x = \frac{75}{4} \\ \therefore F = \frac{4}{3}x & = 18\frac{3}{4} \end{array}$$

OR

$$k = \frac{F}{x}, k \text{ is a constant}$$

$$\frac{20}{15} = \frac{25}{x}, \quad x = \frac{25 \times 15}{20} = 18\frac{3}{4}$$

Class Work (Tier B)

3. Given that p is directly proportional to q^3 , it is known that $p = 24$ for a particular value of q . Find the value of p when this value of q is halved. (N04/PI/15)

$$\begin{array}{l|l} q_1 & k = \frac{p}{q^3}, k \text{ is a constant} \\ \text{Sub } p = 24, q = q_1 & k = \frac{24}{q_1^3} \\ \text{Since } p = kq^3 & q_1 \rightarrow \frac{q_1}{2} \text{ or } \frac{1}{2}q_1 \\ \therefore p = \left(\frac{24}{q_1^3}\right)q^3 & \text{Sub } q = \frac{q_1}{2} \\ & p = \left(\frac{24}{q_1^3}\right)\left(\frac{q_1}{2}\right)^3 \\ & = \left(\frac{24}{q_1^3}\right)\left(\frac{q_1^3}{8}\right) \\ & = 3 \end{array}$$

OR $k = \frac{p}{q^3}, k \text{ is a constant}$

$$\begin{aligned} \frac{24}{q_1^3} &= \frac{p}{\left(\frac{1}{2}q_1\right)^3} \\ p &= \frac{24 \times \frac{1}{8}q_1^3}{q_1^3} = 3 \end{aligned}$$

Class Work (Tier C)

4. An open cylinder has a circular base of variable area. The radius of the circular base is directly proportional to the height of the cylinder. When the volume of the cylinder is $375\pi \text{ cm}^3$, the radius is found to be 5 cm.

- Write down the relationship between radius and height as a formula.
 - Write down the volume of the cylinder in terms of r .
 - Find the total surface area in terms of r .
- (Assumption: negligible thickness)

(a) Let the radius be r and height be h . (b) $\text{Vol} = \pi r^2 h$.

$\therefore k = \frac{h}{r}$, k is constant. we have r , need to find h . Sub $h = 3r$,

$$\begin{aligned} \text{Vol. of open cylinder} &= \pi r^2 h \\ 375\pi &= \pi(5)^2 h \end{aligned}$$

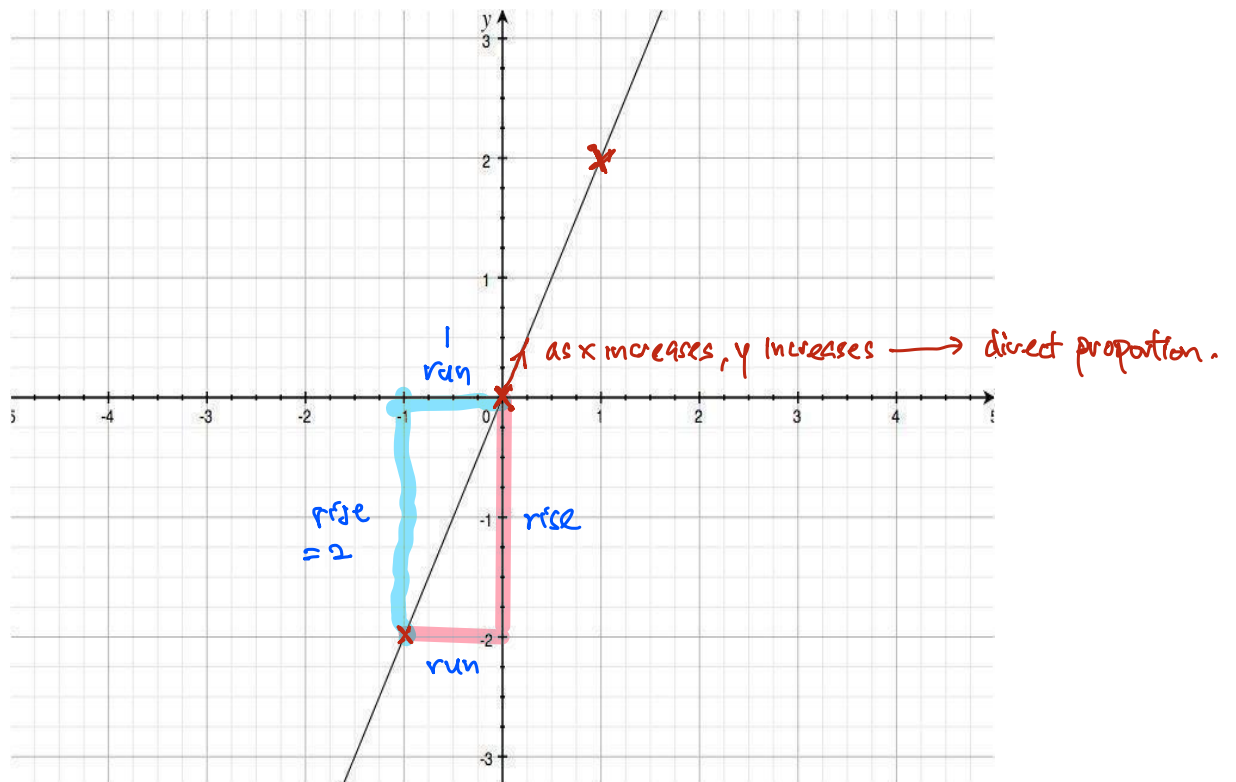
$$\begin{aligned} h &= \frac{375\pi}{25\pi} \\ &= 15 \end{aligned}$$

$$\begin{aligned} h &= kr \quad \therefore k = \frac{15}{5} = 3 \\ \therefore h &= 3r \end{aligned}$$

$$\begin{aligned} &= \pi r^2 (3r) \\ &= 3\pi r^3 \end{aligned}$$

$$\begin{aligned} \text{(c) SA of open cylinder} &= \text{base} + \text{curved SA.} \\ &= \pi r^2 + 2\pi r h. \\ \text{Sub } h &= 3r \\ &= \pi r^2 + 2\pi r (3r) \\ &= \pi r^2 + 6\pi r^2 \\ &= 7\pi r^2 \end{aligned}$$

5. The graph shows a linear relationship.



Find the equation of the line in terms of x and y .

$$\begin{aligned} y &= mx \\ m &= \downarrow \text{gradient} \rightarrow \frac{\text{rise}}{\text{run}} \\ &= \frac{2}{1} \\ &= 2 \\ y &= 2x \end{aligned}$$

6.2 INVERSE PROPORTION



: Access this [Excel](#) document to plot the dynamic data under the “Inverse” Tab

If 5 men take 6 hours to complete a task,

- How many hours would 3 men take to complete the same task?
- How many men would be needed to complete the task in 15 hours?
- Complete the following table.

No. of men, m	1	2	3	4	5
No. of hours, h	30	15	10	7½	6
No. of man-hours, mh	30	30	30	30	30

↑ as m increases,
↓ h decreases
↓
inverse proportion.

- From the above table, what do you notice about h when m increases?

when m increases, h decreases.

- What do you notice about the product mh ?

The rate is a constant.

- Formulate an equation connecting m and h .

$$30 = mh \quad \text{or} \quad m = \frac{30}{h} \quad \text{or} \quad h = \frac{30}{m}$$

fixed constant

use this for equation connecting m & h .

When one quantity increases while the second decreases at the same rate as

another, the two are said to be in inverse proportion.

Memorise (1) Recognise whether it is direct/inverse proportion.

Hence, we have:

(2)

If y is inversely proportional to x , then $xy = k$ or $y = \frac{k}{x}$, where k is a constant and $k \neq 0$.

If y is directly proportional to x , $k = \frac{y}{x}$ or $y = kx$

or $k = \frac{x}{y}$ or $x = ky$ where k is a constant, $k \neq 0$.

Worked Example from Textbook

Worked
Example



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Equation of inverse proportion

If y is inversely proportional to x and $y = 3$ when $x = 4$, find

- (i) the value of y when $x = 8$,
- (ii) an equation connecting x and y ,
- (iii) the value of x when $y = 48$.

Solution

- (i) When $x = 8$, x is doubled from $x = 4$
 $y = \frac{3}{2}$ y is halved from $y = 3$

Alternatively, $x_2 y_2 = x_1 y_1$
 $8 \times y = 4 \times 3$
 $y = \frac{12}{8}$
 $= \frac{3}{2}$

- (ii) Since y is inversely proportional to x ,
then $y = \frac{k}{x}$, where k is a constant.

When $x = 4$, $y = 3$,

$$3 = \frac{k}{4}$$
$$\therefore k = 12$$
$$\therefore y = \frac{12}{x}$$

- (iii) When $y = 48$,

$$48 = \frac{12}{x}$$
$$\therefore x = \frac{12}{48}$$
$$= \frac{1}{4}$$

Attention

- (ii) Since y is inversely proportional to x , then
 $xy = k$ or $y = \frac{k}{x}$, where k is a constant and $k \neq 0$.

Example 1:

Two variables a and b are inversely proportional to each other. If $a = 3$ when $b = 27$, find the value of a when $b = 54$.

$$k = ab, \text{ } k \text{ is a constant.}$$

$$= 3(27)$$

$$= 81.$$

$$a = \frac{81}{b} \quad \text{or} \quad b = \frac{81}{a}.$$

when $b = 54$

$$a = \frac{81}{54}$$

$$= 1.5.$$

OR

$$k = ab, \text{ } k \text{ is a constant.}$$

$$3(27) = a(54)$$

$$a = \frac{3(27)}{54}$$

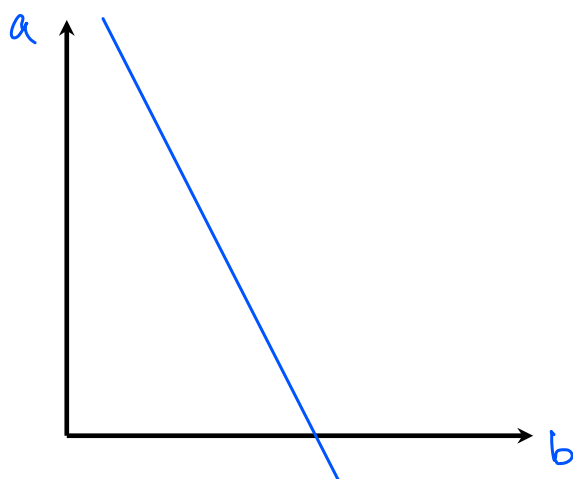
$$= 1.5.$$

Extension:

What do you think the graph will look like? Draw a preliminary sketch.

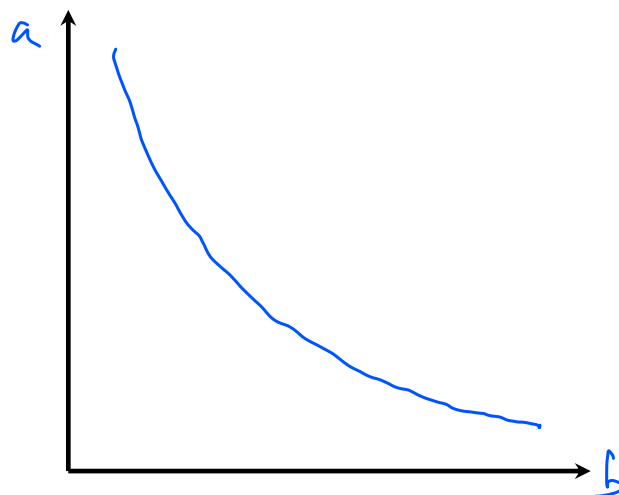
Confirm your prediction using graphing software.

Your prediction:



$$a = \frac{81}{b}.$$

The actual:



Example 2:

Given y is inversely proportional to x^2 , it is known that $y = 4$ when $x = 3$. Find the value of y when $x = 10$. (N04/P1/15)

① $k = y(x^2)$, k is a constant

$$= 4(3^2)$$

$$= 36$$

$$y = \frac{k}{x^2}$$

or $x^2 = \frac{k}{y}$

$$x^2 = \frac{36}{y}$$

② $y = \frac{36}{x^2}$

③ when $x = 10$, $y = \frac{36}{10^2} = \frac{36}{100} = 0.36$ or $\frac{9}{25}$

OR

$$k = y(x^2), k \text{ is a constant}$$

$$4(3^2) = y(10^2)$$

$$y = 0.36$$

Example 3:

The number of days d required to renovate a house is inversely proportional to the number of men available, n . When 6 men are doing the job, the renovation takes 8 days. If it takes 12 days to complete the job, how many men are doing the job?

1 step to get answer!

d	8	12	
n	6	?	

$\downarrow$ $\downarrow$
 $k = 48$ 48

$$k = dn, k \text{ is a constant}$$

$$= 8(6)$$

$$= 48$$

Constant

$$\therefore 12 \times 4 \text{ men} = 48$$

$$\therefore 4 \text{ men}$$

Class Work (Tier A)

1. The frequency, f of a radio wave is inversely proportional to its wavelength, w .

Given that $w = 3000$ m and $f = 100$ kHz, find the frequency when the wavelength is 500 m. Find the wavelength when the frequency is 800 kHz.

② $k = fw$, k is a constant
 $= 100(3000)$
 $= 300000$

③ $f = \frac{k}{w}$
 $f = \frac{300000}{w}$

when $w = 500$
 $f = \frac{300000}{500}$
 $= 600$ kHz.

when $f = 800$
 $800 = \frac{300000}{w}$
 $w = \frac{300000}{800}$
 $= 375$ m

OR

$k = fw$, k is a constant

$100(3000) = 800w$

$w = 375$ m

Class Work (Tier B)

2. The force of attraction, F newtons, between two magnets is inversely proportional to the square of the distance, x centimetres, between them.

(a) When the magnets are 4 cm apart, the force is 3 newtons. Find the formula connecting F and x .

(b) Find the force when the magnets are 2 cm apart.

(c) When the magnets are a certain distance x_1 apart, the force is 1.25 newtons.

Write down the force when the distance is halved.

(N07/P1/19)

(a) $k = Fx^2$, k is a constant
 $= 3(4)^2$
 $= 48$
 $F = \frac{k}{x^2}$
 $F = \frac{48}{x^2}$

(b) when $x = 2$,
 $F = \frac{48}{(2)^2}$
 $= 12$ N

when $x = x_1$, $F = 1.25$.
 $1.25 = \frac{48}{x_1^2} \Rightarrow x_1^2 = \frac{48}{1.25}$
 $= 38.4$
 $x_1 = \frac{x_1}{2}$
 $F = \frac{48}{\left(\frac{x_1}{2}\right)^2}$
 $= \frac{48}{\left(\frac{x_1^2}{4}\right)}$
 $= \frac{48 \times 4}{38.4}$
 $= 5$ N.

Class Work (Tier C)

3. The force, F , between two particles is inversely proportional to the square of the distance, d , between them. The distance between two particles is increased by 25%. Find the percentage change in the force between the particles, stating if it is an increase or a decrease.

$$\begin{array}{lcl} d_1 & \longrightarrow & 1.25d_1 \text{ or } \frac{5}{4}d_1 \\ F_1 & \longrightarrow & F_2 \end{array}$$

$$k = F_1(d_1)^2, \text{ } k \text{ is constant.}$$

$$k = F_2\left(\frac{5}{4}d_1\right)^2$$

$$F_1 d_1^2 = F_2 \left(\frac{25}{16} d_1^2\right)$$

$$F_2 = \frac{F_1 d_1^2}{\frac{25}{16} d_1^2}$$

$$F_2 = \frac{16}{25} F_1 \quad \text{--- Since } F_2 < F_1 \text{ it is a decrease}$$

$$\begin{aligned} \text{Percentage change} &= \frac{9}{25} \times 100\% \\ &= 36\% \end{aligned}$$

It is a decrease.